

RESEARCHES ON THE PRIME DIVISORS OF A CLASS OF FORMULAE OF THE FOURTH DEGREE.

1st Memoir.

In the literary announcements of Göttingen (11 April 1825) one finds the extract of a memoir of indeterminate analysis which Mr. GAUSS presented to the Royal Society of Sciences of that city, but which has not yet been printed. This memoir is the first of a series of memoirs which the illustrious author of the *Disquisitiones arithmeticae* proposes to give on the theory of biquadratic residues, and its object is to determine the distinctive characters of the prime divisors of the formula $x^4 - 2$. The author establishes in it two extremely elegant theorems which can serve to decide whether a prime number, divisor of $x^2 - 2$, divides or does not divide the preceding formula. Having learned, in the course of the year just ended, of the cited extract, which contains only the statements of the two theorems just mentioned and of some auxiliary propositions, I wished to demonstrate on my side the beautiful theorems discovered by Mr. GAUSS. The researches which I made with this aim led me to find a proof founded on extremely simple considerations and probably entirely different from that of Mr. GAUSS, which appears to require very delicate and rather extensive preliminary researches. I then applied analogous considerations to other questions and particularly to the search for the properties which distinguish the prime divisors of the formula $\alpha x^4 + \beta x^2 + \gamma$; and I thus arrived at a large number of interesting theorems. The rapid exposition of part of the results to which these researches have led me is the object of the present memoir. I begin by setting down some definitions and by stating some theorems very easy to establish and on which we shall have to rely in what follows.

1.

“If one can assign to the indeterminate x a value such that $x^4 - A$ becomes divisible by B , A will be called a biquadratic residue with respect to B .”

It is easy to see that, if a number A is a biquadratic residue with respect to a number B , or in other words, if B is a divisor of $x^4 - A$, each prime factor of B will likewise be a divisor of $x^4 - A$, and reciprocally, that if this condition holds with respect to every prime factor of the number B , B will itself be a divisor of $x^4 - A$. Thus, when it is a question of assigning all the numbers which divide the formula $x^4 - A$, one can confine oneself to considering prime numbers.

It is no less evident that, for a number to divide the formula $x^4 - A$, it is necessary that this number be a divisor of $x^2 - A$. There are therefore only the prime divisors of this latter formula to examine, that is to say, the prime numbers with respect to which A is a quadratic residue. One may add that, relative to those among these latter which are of the form $4n + 3$, the question presents no difficulty; for one verifies by a very simple reasoning that every prime number $4n + 3$, divisor of $x^2 - A$, also divides the formula $x^4 - A$.

Let p be a prime number $4n + 1$, divisor of $x^2 - A$, A designating a positive or negative number not divisible by p ; one has, as is known, $A^{\frac{p-1}{2}} \equiv 1 \pmod{p}$. From this one concludes $A^{\frac{p-1}{4}} \equiv \pm 1 \pmod{p}$ and one proves easily that the upper or lower sign will hold according as p divides or does not divide the formula $x^4 - A$. Thus one may state this theorem:

“ A designating a number which is a quadratic residue with respect to the prime number $p = 4n + 1$, one will have $A^{\frac{p-1}{4}} \equiv 1$ or $A^{\frac{p-1}{4}} \equiv -1 \pmod{p}$. In the first case, A will be a biquadratic residue with respect to p , in the second, A will be a biquadratic non-residue with respect to p .”

If this theorem is applied to the case where $A = -1$, one will find that -1 is a biquadratic residue with respect to the prime numbers $8n + 1$, and on the contrary a non-residue relative to those of the form $8n + 5$. From the preceding theorem one easily deduces the following other theorem, in whose statement one supposes, as in all that follows, that p is a prime number $4n + 1$, and that A and A' designate numbers not divisible by p and which are both quadratic residues with respect to p .

“If the numbers A and A' are both biquadratic residues or both biquadratic non-residues with respect to p , the product AA' will be a biquadratic residue with respect to p ; if, on the contrary, one of the numbers A and A' is a residue and the other a biquadratic non-residue relative to p , AA' will be a biquadratic non-residue with respect to p .”

Taking $A' = -1$, and having regard to what precedes, one will see that, if p is of the form $8n + 1$, A will at the same time as $-A$ be a biquadratic residue or non-residue, and that, on the contrary, if p is of the form $8n + 5$, one of the numbers A and $-A$ will be a residue and the other a biquadratic non-residue with respect to p .

2.

After establishing these preliminaries, we shall deal with the search for the characters which distinguish the prime divisors of the formula $x^4 - 2$. It is known that the prime divisors of $x^2 - 2$ are of one of these two forms: $8n + 1$, $8n + 7$, and that reciprocally every prime number of one of these forms divides the formula $x^2 - 2$. According to what we said in the preceding paragraph, we need not take account of the latter of these two forms, and it will suffice to consider the prime numbers $8n + 1$. Let p be a prime number of this kind, and put, as one is permitted to do, $p = t^2 + 2u^2$, where u will be even and t odd. Let $u = 2^\nu k k' k'' \dots$, 2^ν being the highest power of 2 which divides u , and $k, k', k'', \dots$ designating odd prime numbers, several of which may be equal to one another. The equation $t^2 + 2u^2 = p$ immediately gives $t^2 \equiv p \pmod{k}$. The number p is therefore a quadratic residue with respect to k , which we shall write thus: $\left(\frac{p}{k}\right) = 1$, adopting the notation used by Mr. LEGENDRE. Recall that, if c designates a prime number and M any number not divisible by c , this illustrious geometer uses the sign $\left(\frac{M}{c}\right)$ to designate the remainder which one obtains by dividing by c the power $M^{\frac{c-1}{2}}$, a remainder which is known to be equal to 1 or -1 , according as M is or is not a quadratic residue with respect to c . Applying to the relation $\left(\frac{p}{k}\right) = 1$ the theorem known under the name law of reciprocity (the *theorema fundamentale* of Mr. GAUSS), one will have, p being

of the form $4n + 1$, $\left(\frac{k}{p}\right) = 1$. One finds in the same way:

$$\left(\frac{k'}{p}\right) = 1, \quad \left(\frac{k''}{p}\right) = 1, \quad \text{etc.}$$

On the other hand, since p is of the form $8n + 1$, one also has $\left(\frac{2}{p}\right) = 1$, and consequently $\left(\frac{2^\nu}{p}\right) = 1$. Multiplying this last relation by all the preceding ones, one obtains:

$$\left(\frac{2^\nu k k' k'' \dots}{p}\right) = \left(\frac{u}{p}\right) = 1.$$

Now consider the simple factors of the odd number t , which we shall divide into two classes. The first class will comprise the prime divisors of one of these two forms: $8n + 1$, $8n + 7$, and the numbers belonging to it will be designated by $g, g', g'', \dots$; the second class will be composed of numbers $h, h', h'', \dots$, contained in these two forms: $8n + 3$, $8n + 5$. One first has:

$$t = gg'g'' \dots \times hh'h'' \dots,$$

and one will then conclude from the equation $p = t^2 + 2u^2$:

$$\begin{aligned} \left(\frac{2p}{g}\right) &= 1, & \left(\frac{2p}{g'}\right) &= 1, & \left(\frac{2p}{g''}\right) &= 1, & \text{etc.} \\ \left(\frac{2p}{h}\right) &= 1, & \left(\frac{2p}{h'}\right) &= 1, & \left(\frac{2p}{h''}\right) &= 1, & \text{etc.} \end{aligned}$$

On the other hand, by known theorems one has:

$$\begin{aligned} \left(\frac{2}{g}\right) &= 1, & \left(\frac{2}{g'}\right) &= 1, & \left(\frac{2}{g''}\right) &= 1, & \text{etc.} \\ \left(\frac{2}{h}\right) &= -1, & \left(\frac{2}{h'}\right) &= -1, & \left(\frac{2}{h''}\right) &= -1, & \text{etc.} \end{aligned}$$

Comparing these relations with the preceding ones, one finds:

$$\begin{aligned} \left(\frac{p}{g}\right) &= 1, & \left(\frac{p}{g'}\right) &= 1, & \left(\frac{p}{g''}\right) &= 1, & \text{etc.} \\ \left(\frac{p}{h}\right) &= -1, & \left(\frac{p}{h'}\right) &= -1, & \left(\frac{p}{h''}\right) &= -1, & \text{etc.} \end{aligned}$$

The application of the law of reciprocity to these last relations gives:

$$\begin{aligned} \left(\frac{g}{p}\right) &= 1, & \left(\frac{g'}{p}\right) &= 1, & \left(\frac{g''}{p}\right) &= 1, & \text{etc.} \\ \left(\frac{h}{p}\right) &= -1, & \left(\frac{h'}{p}\right) &= -1, & \left(\frac{h''}{p}\right) &= -1, & \text{etc.} \end{aligned}$$

hence, on multiplying:

$$\left(\frac{gg'g'' \dots \times hh'h'' \dots}{p} \right) = \left(\frac{t}{p} \right) = \pm 1,$$

where the upper or lower sign must be taken according as the numbers $h, h', h'', \dots$ are even or odd in number. But it is easy to see, from the equation $t = gg'g'' \dots \times hh'h'' \dots$, that the first case will occur when t is of one of these forms: $8n + 1, 8n + 7$, and the second when t is contained in one of these: $8n + 3, 8n + 5$. Thus one has:

$$\begin{aligned} \left(\frac{t}{p} \right) &= 1, & \text{when } t &= 8n + 1 \quad \text{or} \quad 8n + 7, \\ \left(\frac{t}{p} \right) &= -1, & \text{when } t &= 8n + 3 \quad \text{or} \quad 8n + 5. \end{aligned}$$

Return to the equation $t^2 + 2u^2 = p$ and put it in the form of a congruence:

$$t^2 \equiv -2u^2 \pmod{p}.$$

Raising both sides to the power $\frac{p-1}{4}$, one will have $\left(\frac{p-1}{4} \right.$ being even):

$$t^{\frac{p-1}{2}} \equiv 2^{\frac{p-1}{4}} u^{\frac{p-1}{2}} \pmod{p},$$

or, what comes to the same thing, having proved that $\left(\frac{u}{p} \right) = 1$:

$$t^{\frac{p-1}{2}} \equiv 2^{\frac{p-1}{4}} \pmod{p}.$$

Thus one sees that ± 2 (the double sign may be put, since $p = 8n + 1$) is or is not a biquadratic residue with respect to p according as one has $\left(\frac{t}{p} \right) = 1$ or $\left(\frac{t}{p} \right) = -1$. Comparing this result with what precedes, one has the theorem:

“ p designating a prime number $8n + 1$, if one writes $p = t^2 + 2u^2$, I say that ± 2 will or will not be a biquadratic residue with respect to p according as t is of one of these forms: $8n + 1, 8n + 7$ or of one of these: $8n + 3, 8n + 5$.”

This is the first of the two theorems of Mr. GAUSS mentioned in the preamble of this memoir. The second of these theorems is relative to the decomposition of the number p into two squares, and can easily be deduced from the one just established.

Let $p = \varphi^2 + \psi^2$ (where ψ is supposed divisible by 4) and equate this value of p with the one we have just considered.

We shall thus have:

$$p = t^2 + 2u^2 = \varphi^2 + \psi^2,$$

and, after transposing:

$$t^2 - \psi^2 = (t + \psi)(t - \psi) = \varphi^2 - 2u^2.$$

Since φ is odd, the greatest common divisor of φ and u will be odd; designate it by m and put $\varphi = m\varphi'$, $u = mu'$. Substitution of these values in the preceding equation changes it into:

$$(t + \psi)(t - \psi) = m^2(\varphi'^2 - 2u'^2).$$

The odd number $t + \psi$ is thus composed of two factors E and K , the first of which divides m^2 , the second $\varphi'^2 - 2u'^2$; and the same is true of $t - \psi$, whose factors we denote by F and L , L being allowed to be negative. We therefore have the equations:

$$\begin{aligned} t + \psi &= EK, & t - \psi &= FL, \\ m^2 &= EF, & \varphi'^2 - 2u'^2 &= KL. \end{aligned}$$

It is easy to verify that the numbers E and F are relatively prime. Indeed, let δ be a prime divisor of E , and suppose that δ also divides F . The number δ would be a common divisor of $t + \psi$ and $t - \psi$, and consequently would divide the number t , which is the half-sum of the preceding two numbers. On the other hand, from the fact that δ is a prime divisor of E , it follows successively, having regard to the equations $EF = m^2$, $u = mu'$, that m^2 , m , and u are multiples of δ . The numbers t and u would therefore have the common divisor δ , and $p = t^2 + 2u^2$ would not be a prime number.

The product of the numbers E and F , which are relatively prime, being a square, each of them is also a square. Let $E = e^2$; then $t + \psi = e^2K$. The odd square e^2 is of the form $8n + 1$, and K , as an odd divisor of $\varphi'^2 + 2u'^2$ (where φ' and u' are relatively prime), is of one of these forms: $8n + 1$, $8n + 7$. Thus the number $t + \psi$ will itself be of one of the forms $8n + 1$, $8n + 7$. The number ψ , which we know to be divisible by 4, will be of the form $8n$ or of the form $8n + 4$. It follows from the preceding that, in the first case, t will be of one of these two forms: $8n + 1$, $8n + 7$, and in the second of one of these: $8n + 3$, $8n + 5$. Comparing this result with the preceding theorem, one has this other theorem:

“ p designating a prime number $8n + 1$, and having written $p = \varphi^2 + \psi^2$ (where ψ is supposed divisible by 4), ± 2 will or will not be a biquadratic residue with respect to p according as ψ is of the form $8n$ or of the form $8n + 4$.”

There is a third theorem suitable for deciding whether ± 2 is or is not a biquadratic residue relative to a prime number $p = 8n + 1$, which may be stated as follows:

“Having made, in any manner, $p = t^2 - 2u^2$, ± 2 will or will not be a biquadratic residue with respect to p according as t is of one of the forms $8n + 1$, $8n + 3$, or of one of these: $8n + 5$, $8n + 7$.”

We shall not stop to prove this theorem, which can be established directly and by considerations analogous to those on which the proof of the first of the two preceding theorems is founded. One can also deduce it from each of the preceding theorems almost as one has just passed from the first to the second.

3.

We now pass to more general considerations. Let b be a prime number $4n + 3$, p a prime number $4n + 1$ capable of being put under the form $t^2 - bu^2$, and let us propose to decide whether $-b$ is or is not a biquadratic residue with respect to p . It is easy to see that, if p can be put under the form $t^2 - bu^2$, one can always do so in such a way that u is even, and consequently t odd^{*)}. Let therefore $p = t^2 - bu^2$, and put $u = 2^\nu k k' k'' \dots$, where $k, k', k'', \dots$ are the simple odd factors of u . One concludes immediately from the preceding equation:

$$\left(\frac{p}{k}\right) = 1, \quad \left(\frac{p}{k'}\right) = 1, \quad \left(\frac{p}{k''}\right) = 1, \quad \text{etc.}$$

The application of the reciprocity law then gives, p being of the form $4n + 1$:

$$\left(\frac{k}{p}\right) = 1, \quad \left(\frac{k'}{p}\right) = 1, \quad \left(\frac{k''}{p}\right) = 1, \quad \text{etc.}$$

Multiplying these relations together and with the identical relation $\left(\frac{2^\nu}{p}\right) = \left(\frac{2^\nu}{p}\right)$,

^{*)}To prove that this is always possible, we shall show that, if one has $t^2 - bu^2 = p$, t being even, it is easy to deduce from the values of t and u other numbers t' (odd) and u' which likewise satisfy the equation $t'^2 - bu'^2 = p$. Let r and s be the least numbers such that $r^2 - bs^2 = 1$; I say that r will be even. Indeed, it is known that, if b designates a prime number $4n + 3$, the equation $\rho^2 - b\sigma^2 = \pm 2$ is always possible, and that, ρ and σ being supposed the least numbers which satisfy it, one has $r = b\sigma^2 \pm 1$, $s = \rho\sigma$ (*Theory of Numbers*, no. 44.45). From this, and from the fact that the numbers ρ and σ are evidently both odd, it follows that r is an even number. This being so, if one multiplies together the equations $r^2 - bs^2 = 1$, $t^2 - bu^2 = p$, one obtains $(rt \pm bsu)^2 - b(ru \pm st)^2 = p$, and one will easily see that $rt \pm bsu$ is an odd number.

one has the result:

$$(\alpha) \quad \left(\frac{u}{p}\right) = \left(\frac{2^\nu}{p}\right).$$

Now decompose the odd number t into its simple factors and divide these factors into two classes. Those of the first class will be designated by $g, g', g'', \dots$, and are such that:

$$(\beta) \quad \left(\frac{-b}{g}\right) = 1, \quad \left(\frac{-b}{g'}\right) = 1, \quad \left(\frac{-b}{g''}\right) = 1, \quad \text{etc.}$$

As for those which form the second class and which we shall designate by $h, h', h'', \dots$, they are such that:

$$(\beta') \quad \left(\frac{-b}{h}\right) = -1, \quad \left(\frac{-b}{h'}\right) = -1, \quad \left(\frac{-b}{h''}\right) = -1, \quad \text{etc.}$$

The product of all these numbers is equal to t , that is:

$$t = gg'g'' \dots \times hh'h'' \dots$$

Inspection of the equation $t^2 - bu^2 = p$ gives the results:

$$\begin{aligned} \left(\frac{-bp}{g}\right) &= 1, \quad \left(\frac{-bp}{g'}\right) = 1, \quad \left(\frac{-bp}{g''}\right) = 1, \quad \text{etc.} \\ \left(\frac{-bp}{h}\right) &= 1, \quad \left(\frac{-bp}{h'}\right) = 1, \quad \left(\frac{-bp}{h''}\right) = 1, \quad \text{etc.} \end{aligned}$$

Comparison of these relations with the preceding ones then gives:

$$\begin{aligned} \left(\frac{p}{g}\right) &= 1, \quad \left(\frac{p}{g'}\right) = 1, \quad \left(\frac{p}{g''}\right) = 1, \quad \text{etc.} \\ \left(\frac{p}{h}\right) &= -1, \quad \left(\frac{p}{h'}\right) = -1, \quad \left(\frac{p}{h''}\right) = -1, \quad \text{etc.} \end{aligned}$$

Applying now the reciprocity law, one obtains:

$$\begin{aligned} \left(\frac{g}{p}\right) &= 1, \quad \left(\frac{g'}{p}\right) = 1, \quad \left(\frac{g''}{p}\right) = 1, \quad \text{etc.} \\ \left(\frac{h}{p}\right) &= -1, \quad \left(\frac{h'}{p}\right) = -1, \quad \left(\frac{h''}{p}\right) = -1, \quad \text{etc.} \end{aligned}$$

Multiplying these relations together, one will have:

$$\left(\frac{gg'g'' \dots \times hh'h'' \dots}{p}\right) = \left(\frac{t}{p}\right) = \pm 1,$$

the upper or lower sign holding according as the numbers $h, h', h'', \dots$ are even or odd in number. On the other hand, if one applies the reciprocity law to the relations (β) and (β') , one obtains, b being of the form $4n + 3$:

$$\begin{aligned} \left(\frac{g}{b}\right) = 1, \quad \left(\frac{g'}{b}\right) = 1, \quad \left(\frac{g''}{b}\right) = 1, \quad \text{etc.} \\ \left(\frac{h}{b}\right) = -1, \quad \left(\frac{h'}{b}\right) = -1, \quad \left(\frac{h''}{b}\right) = -1, \quad \text{etc.} \end{aligned}$$

equalities which, when multiplied together, give:

$$\left(\frac{gg'g'' \dots \times hh'h'' \dots}{b}\right) = \left(\frac{t}{b}\right) = \pm 1,$$

where the upper or lower sign must be taken according as the numbers $h, h', h'', \dots$ are even or odd in number.

Comparison of this result with the one obtained a moment ago shows that one always has:

$$(\gamma) \quad \left(\frac{t}{p}\right) = \left(\frac{t}{b}\right).$$

Let us now return to the equation $p = t^2 - bu^2$ and put it in congruence form:

$$t^2 \equiv bu^2 \pmod{p}.$$

From this, raising both sides to the power $\frac{p-1}{4}$, one obtains:

$$t^{\frac{p-1}{2}} \equiv b^{\frac{p-1}{4}} u^{\frac{p-1}{2}} \pmod{p}.$$

The formula (α) , namely $\left(\frac{u}{p}\right) = \left(\frac{2^\nu}{p}\right)$, where 2^ν denotes the highest power dividing u , can be presented in another manner. For this it is necessary to distinguish two cases, according as p is of the form $8n+1$ or of the form $8n+5$. If p is a prime number $8n+1$, one has, as is known, $\left(\frac{2}{p}\right) = 1$, and consequently $\left(\frac{2^\nu}{p}\right) = 1$; the formula in question therefore becomes in this case $\left(\frac{u}{p}\right) = 1$. If p is a prime number $8n+5$, the number ν is $= 1$; for if ν were greater than unity, u^2 would be divisible by 8, and $p = t^2 - bu^2$ would be of the form $8n+1$. Moreover, since in this case $\left(\frac{2}{p}\right) = -1$, formula (α) becomes $\left(\frac{u}{p}\right) = -1$. The two cases just examined are included in the formula $\left(\frac{u}{p}\right) = (-1)^{\frac{p-1}{4}}$, which is equivalent to the congruence $u^{\frac{p-1}{2}} \equiv (-1)^{\frac{p-1}{4}} \pmod{p}$. Substituting the value which this gives

for $u^{\frac{p-1}{2}}$ in the congruence obtained above, one has:

$$t^{\frac{p-1}{2}} \equiv (-b)^{\frac{p-1}{4}} \pmod{p},$$

a result which shows that $-b$ will or will not be a biquadratic residue with respect to p according as t is or is not a quadratic residue with respect to p . Comparing this result with that contained in formula (γ), one arrives at the theorem which we shall state:

“Let b designate a prime number $4n + 3$, and p a prime number $4n + 1$ capable of being put under the form $t^2 - bu^2$. Having written $p = t^2 - bu^2$ (where t is supposed odd), I say that $-b$ will or will not be a biquadratic residue with respect to p according as t is or is not a quadratic residue with respect to b .”

4.

We shall now deduce from this theorem another by means of which one can decide still more quickly whether $-b$ is or is not a biquadratic residue with respect to p . Preserve the preceding notations and put $p = \varphi^2 + \psi^2$ (where ψ is supposed even). Equating this value of p to the one considered above, one has:

$$p = t^2 - bu^2 = \varphi^2 + \psi^2,$$

and, after transposing:

$$t^2 - \psi^2 = (t + \psi)(t - \psi) = \varphi^2 + bu^2.$$

There are now two cases to distinguish, according as φ is or is not divisible by b . We begin with the latter of these two cases. Let m be the greatest common divisor of φ and u , which will be odd (φ being odd) and not divisible by b , and put $\varphi = m\varphi'$, $u = mu'$. Substitution of these values in the last equation changes it into:

$$(t + \psi)(t - \psi) = m^2(\varphi'^2 + bu'^2).$$

It is evident, from this equation, that $t + \psi$ is composed of two factors E and K , the first of which divides m^2 , the second $\varphi'^2 + bu'^2$, and that the same is true of $t - \psi$. Denoting the factors of this last number by F and L , we shall have the equations:

$$\begin{aligned} t + \psi &= EK, & t - \psi &= FL, \\ m^2 &= EF, & \varphi'^2 + bu'^2 &= KL. \end{aligned}$$

It is very easy to make sure that the numbers E and F are relatively prime. Indeed, let δ be any simple factor of E (which will necessarily be odd, E itself being odd) and suppose that δ also divides F . Inspection of the preceding equations shows that, under this supposition, δ would be a common divisor of $t + \psi$ and $t - \psi$, and consequently would divide the number t , which is the half-sum of the preceding two. But on the other hand one sees that δ , as a prime divisor of E , divides m^2 , and consequently m and u (u being $= mu'$). The numbers t and u would therefore, under this supposition, have the common factor δ , which is absurd, since $p = t^2 - bu^2$ is a prime number. It is therefore proved that E and F cannot have a common divisor, and since the product of these numbers is a square, each of them is likewise a square. Thus one has:

$$\left(\frac{E}{b}\right) = 1, \quad \left(\frac{F}{b}\right) = 1.$$

As for the numbers K and L , they are likewise such that one has:

$$\left(\frac{K}{b}\right) = 1, \quad \left(\frac{L}{b}\right) = 1.$$

Indeed, one sees from the last equation (p. 74) that these numbers divide, the one and the other, the formula $\varphi'^2 + bu'^2$ (where φ' and bu' are relatively prime) and one knows by a known theorem that follows easily from the law of reciprocity (*Theory of Numbers*, no. 197) that every odd divisor R of such a formula is such that $\left(\frac{R}{b}\right) = 1$. Multiplying the first of the last relations by the third, and the second by the fourth, one obtains:

$$\left(\frac{EK}{b}\right) = 1, \quad \left(\frac{FL}{b}\right) = 1,$$

or, which is the same thing:

$$(\delta) \quad \left(\frac{t + \psi}{b}\right) = 1, \quad \left(\frac{t - \psi}{b}\right) = 1.$$

Examining in a similar way the case in which φ is divisible by b , one will find that, in this case, one of the numbers $t + \psi$ and $t - \psi$ is divisible by b , and that the other is, as in the case with which we have just been occupied, a quadratic residue with respect to b , so that, denoting by $t \pm \psi$ that one of these numbers which is divisible by b , one has:

$$(\delta') \quad t \pm \psi \equiv 0 \pmod{b}, \quad \left(\frac{t \mp \psi}{b}\right) = 1.$$

It follows from the theorem stated at the end of the preceding paragraph that, in order to decide whether $-b$ is or is not a biquadratic residue with respect to p , everything reduces to the question of knowing whether t is or is not a quadratic residue with respect to b . We can now show, by means of the results just obtained, that this latter question can be decided without knowing t , that is, without its being necessary to put p under the form $t^2 - bu^2$. Suppose first that φ is not divisible by b . The relations (δ) , which hold in this case, can be presented in this way, observing that b is a prime number $4n + 3$:

$$\left(\frac{\psi + t}{b}\right) = 1, \quad \left(\frac{\psi - t}{b}\right) = -1.$$

One immediately draws from the equation $p = t^2 - bu^2$: $t^2 \equiv p \pmod{b}$, a result which shows that, on solving the congruence $\chi^2 \equiv p \pmod{b}$, and denoting by χ one of its roots taken at random, one will have $t \equiv \chi$, or $t \equiv -\chi \pmod{b}$. In the first case, one has:

$$\left(\frac{t}{b}\right) = \left(\frac{\chi}{b}\right), \quad \left(\frac{\chi + \psi}{b}\right) = 1,$$

for it is evident that the expressions $\left(\frac{t}{b}\right)$ and $\left(\frac{t+\psi}{b}\right)$ do not change when one puts, in place of t , another number χ which differs from t only by a multiple of b . Multiplying the last relations together, one obtains:

$$\left(\frac{t}{b}\right) = \left(\frac{\chi(\chi + \psi)}{b}\right).$$

Consider now the other case, in which $t \equiv -\chi \pmod{b}$. One will then have:

$$\left(\frac{t}{b}\right) = -\left(\frac{\chi}{b}\right), \quad \left(\frac{\psi + \chi}{b}\right) = -1,$$

the last of these relations resulting from the formula $\left(\frac{\psi-t}{b}\right) = -1$, if one puts, in place of $-t$, the number χ , which differs from it only by a multiple of b . Multiplying the preceding relations together, one will find, as above:

$$\left(\frac{t}{b}\right) = \left(\frac{\chi(\chi + \psi)}{b}\right).$$

We have therefore arrived at this remarkable result: If φ is not divisible by b , one can decide very simply whether one has:

$$\left(\frac{t}{b}\right) = 1 \quad \text{or} \quad \left(\frac{t}{b}\right) = -1.$$

It will suffice to seek a number satisfying the congruence $\chi^2 \equiv p \pmod{b}$; having found such a number χ , one will always have:

$$\left(\frac{t}{b}\right) = \left(\frac{\chi(\chi + \psi)}{b}\right).$$

Let us now come to the case in which φ is divisible by b . The relations (δ') which then hold are:

$$t \pm \psi \equiv 0 \pmod{b}, \quad \left(\frac{t \mp \psi}{b}\right) = 1.$$

In the last of these relations one may add to $t \mp \psi$ any multiple of b . If one adds to it the number $t \pm \psi$, which, by virtue of the first relation, is divisible by b , there will result $\left(\frac{2t}{b}\right) = 1$, a result which entails this other one: $\left(\frac{t}{b}\right) = \left(\frac{2}{b}\right)$.

Combining this result with a known theorem, one will see that, in the case of φ divisible by b , one has $\left(\frac{t}{b}\right) = 1$ or $\left(\frac{t}{b}\right) = -1$, according as t is of the form $8n + 7$ or of this one: $8n + 3$.

Comparing what has just been proved with the theorem established at the end of the preceding paragraph, one will see that it is possible to decide, independently of the knowledge of the number t , whether $-b$ is or is not a biquadratic residue with respect to p . Since the result thus reached no longer contains any trace of the number t , one is led to believe that it does not presuppose the possibility of the equation $t^2 - bu^2 = p$, from which the number t takes its origin, and that it is generally true for all values of b and p such that $\left(\frac{b}{p}\right) = 1$. This is indeed what takes place, as one can prove, by examining, instead of the equation $t^2 - bu^2 = p$, the more general equation $t^2 \pm Mu^2 = p$, where M denotes the product of any number of distinct prime numbers, and then combining the results of this examination with the following theorem, of whose truth one cannot doubt, but whose rigorous demonstration nevertheless presents difficulties:

“ c denoting any prime number whatever and p a prime number $4n + 1$, such that $\left(\frac{c}{p}\right) = 1$, one can always determine a number δ , composed of simple factors all smaller than c , and such that the equation $t^2 \pm c\delta u^2 = p$, is soluble.”

However that may be, one may remark that the theorem which we are about to state is rigorously proved, by what precedes, for all values

of b such that the formula $t^2 - bu^2$ has only the single quadratic divisor $\pm(t^2 - bu^2)$. Casting one's eyes over the first of the tables added to the *Theory of Numbers*, one sees that all the prime numbers $4n + 3$ smaller than 136 (the limit of the table) are in this case, excepting the single number 79. It would be easy, by following the path just indicated, to prove the truth of the theorem for this value or for any other particular value of b , whatever p may be; but in order to establish it in all its generality, it is necessary first to prove, as we have already said, that one can always satisfy the condition which we stated only a moment ago.

Theorem I.

" b denoting a prime number $4n + 3$, and p a prime number $4n + 1$ such that $\left(\frac{b}{p}\right) = 1$, if one puts $p = \varphi^2 + \psi^2$ (where ψ is supposed even) one will have the following rule for deciding whether $-b$ is or is not a biquadratic residue with respect to p : If φ is divisible by b , $-b$ will or will not be a biquadratic residue, according as b is of the form $8n + 7$ or of this one: $8n + 3$. If φ is not divisible by b , one will seek a number χ such that $\chi^2 \equiv p \pmod{b}$. This being done, $-b$ will or will not be a biquadratic residue with respect to p , according as one has $\left(\frac{\chi(\chi+\psi)}{b}\right) = 1$ or $\left(\frac{\chi(\chi+\psi)}{b}\right) = -1$."

By successively taking $b = 3$, $b = 7$ etc., one will obtain the following particular theorems, which are analogous to the second of Mr. GAUSS's theorems and which must be regarded as rigorously proved by the considerations that we have set out in this paragraph and in the preceding one.

" p denoting a prime number $12n + 1$, if one puts $p = \varphi^2 + \psi^2$ (where ψ is supposed even) one of the numbers φ and ψ will necessarily be divisible by 3. This being done, I say that -3 will or will not be a biquadratic residue with respect to p , according as it is ψ or φ which is divisible by 3."

" p denoting a prime number of one of these forms: $28n + 1$, $28n + 9$, $28n + 25$, if one puts $p = \varphi^2 + \psi^2$, I say that -7 will be a biquadratic residue with respect to p , if one of the numbers φ and ψ is divisible by 7, and that -7 will be a biquadratic non-residue with respect to p , if neither of these numbers is divisible by 7."

etc.

5.

Let us denote by a a prime number $4n + 1$, and by p a prime number likewise $4n + 1$, and moreover capable of being put under the form $t^2 - au^2$. We can therefore put $p = t^2 - au^2$ where the number t , as is easy to see, is necessarily odd. One will find, as in paragraph 3, that if one denotes by 2^ν the highest power of 2 that divides u , one has $\left(\frac{u}{p}\right) = \left(\frac{2^\nu}{p}\right)$, a relation which one will then change, as in the cited place, into this one: $\left(\frac{u}{p}\right) = (-1)^{\frac{p-1}{4}}$.

Let us now decompose the odd number t into its simple factors, putting $t = gg'g'' \dots \times hh'h'' \dots$, where the prime numbers $g, g', g'', \text{etc.}, h, h', h'', \text{etc.}$ are such that:

$$(\varepsilon) \quad \begin{cases} \left(\frac{-a}{g}\right) = 1, & \left(\frac{-a}{g'}\right) = 1, & \left(\frac{-a}{g''}\right) = 1, & \text{etc.} \\ \left(\frac{-a}{h}\right) = -1, & \left(\frac{-a}{h'}\right) = -1, & \left(\frac{-a}{h''}\right) = -1, & \text{etc.} \end{cases}$$

The equation $t^2 - au^2 = p$ gives immediately:

$$\begin{aligned} \left(\frac{-ap}{g}\right) &= 1, & \left(\frac{-ap}{g'}\right) &= 1, & \left(\frac{-ap}{g''}\right) &= 1, & \text{etc.} \\ \left(\frac{-ap}{h}\right) &= 1, & \left(\frac{-ap}{h'}\right) &= 1, & \left(\frac{-ap}{h''}\right) &= 1, & \text{etc.} \end{aligned}$$

Comparing these relations with the preceding ones, one obtains:

$$\begin{aligned} \left(\frac{p}{g}\right) &= 1, & \left(\frac{p}{g'}\right) &= 1, & \left(\frac{p}{g''}\right) &= 1, & \text{etc.} \\ \left(\frac{p}{h}\right) &= -1, & \left(\frac{p}{h'}\right) &= -1, & \left(\frac{p}{h''}\right) &= -1, & \text{etc.} \end{aligned}$$

The application of the law of reciprocity will then give, p being of the form $4n + 1$:

$$\begin{aligned} \left(\frac{g}{p}\right) &= 1, & \left(\frac{g'}{p}\right) &= 1, & \left(\frac{g''}{p}\right) &= 1, & \text{etc.} \\ \left(\frac{h}{p}\right) &= -1, & \left(\frac{h'}{p}\right) &= -1, & \left(\frac{h''}{p}\right) &= -1, & \text{etc.} \end{aligned}$$

whence one concludes, by multiplying:

$$\left(\frac{gg'g'' \dots \times hh'h'' \dots}{p}\right) = \left(\frac{t}{p}\right) = \pm 1,$$

the upper or lower sign taking place according as the prime numbers h ,

$h', h'', \dots$ are even or odd in number. One can state this result in a slightly different way, by saying that $\left(\frac{t}{p}\right)$ is equal to the product of the second members of the relations (ε) .

In the relations (ε) , one may everywhere change $-a$ into a , provided that at the same time one changes the sign of the second members of those of these relations in which there occurs a prime number $g, g', g'', \dots, h, h', h'', \dots$ of the form $4n + 3$. One will thus have:

$$(\zeta) \quad \begin{cases} \left(\frac{a}{g}\right) = \pm 1, & \left(\frac{a}{g'}\right) = \pm 1, & \left(\frac{a}{g''}\right) = \pm 1, & \text{etc.} \\ \left(\frac{a}{h}\right) = \pm 1, & \left(\frac{a}{h'}\right) = \pm 1, & \left(\frac{a}{h''}\right) = \pm 1, & \text{etc.} \end{cases}$$

Comparing the second member of each of these relations with the second member of the corresponding relation of table (ε) , one will find as many changes of sign as there are prime numbers $4n + 3$ among the numbers $g, g', g'', \dots, h, h', h'', \dots$.

The number of changes will therefore be even, when among the numbers $g, g', g'', \dots, h, h', h'', \dots$ there is an even number of the form $4n + 3$. In this case, which will occur every time that $t = gg'g'' \dots \times hh'h'' \dots$ is of the form $4n + 1$, the product of the second members of the relations (ζ) will therefore be the same as the product of the second members of the relations (ε) .

On the other hand, one may, by virtue of the law of reciprocity, reverse the first members of the relations (ζ) without its being necessary to change the sign of any of the second members of these relations. The product of all the reversed expressions, that is, $\left(\frac{gg'g'' \dots \times hh'h'' \dots}{a}\right)$ or $\left(\frac{t}{a}\right)$, is therefore equal to the product of the second members of the relations (ζ) and consequently also, according to what has previously been seen, equal to the product of the second members of the relations (ε) . Now, having proved above that this last product is equal to $\left(\frac{t}{p}\right)$, one has:

$$\left(\frac{t}{p}\right) = \left(\frac{t}{a}\right).$$

This result concerns the case in which t is of the form $4n + 1$; if one examines in a similar way the case of $t = 4n + 3$, one will find that one then has:

$$\left(\frac{t}{p}\right) = -\left(\frac{t}{a}\right).$$

These two cases are comprised in the following formula, to which we join a result obtained above:

$$(\eta) \quad \left(\frac{t}{p}\right) = (-1)^{\frac{t-1}{2}} \left(\frac{t}{a}\right), \quad \left(\frac{u}{p}\right) = (-1)^{\frac{p-1}{4}}.$$

The equation $t^2 - au^2 = p$ gives successively:

$$t^2 \equiv au^2, \quad t^{\frac{p-1}{2}} \equiv a^{\frac{p-1}{4}} u^{\frac{p-1}{2}} \pmod{p},$$

whence one sees that a will or will not be a biquadratic residue with respect to p , according as one has $\left(\frac{t}{p}\right) = \left(\frac{u}{p}\right)$ or $\left(\frac{t}{p}\right) = -\left(\frac{u}{p}\right)$. Putting in place of $\left(\frac{t}{p}\right)$ and $\left(\frac{u}{p}\right)$ the values given by the expressions (η) , one can replace the preceding conditions by those that follow:

a will or will not be a biquadratic residue with respect to p , according as one has:

$$(\theta) \quad \left(\frac{t}{a}\right) = (-1)^{\frac{t-1}{2} + \frac{p-1}{4}} \quad \text{or} \quad \left(\frac{t}{a}\right) = -(-1)^{\frac{t-1}{2} + \frac{p-1}{4}},$$

a result that we shall not pause to demonstrate.

Let us now put $p = \varphi^2 + \psi^2$ (where ψ is supposed even) and equate this value of p to the one we considered previously. We shall thus have $p = t^2 - au^2 = \varphi^2 + \psi^2$, whence one concludes by transposing:

$$t^2 - \psi^2 = (t + \psi)(t - \psi) = \varphi^2 + au^2.$$

There are now two cases to distinguish, according as φ is or is not divisible by a . Let us begin with that in which φ is not divisible by a . Let m be the greatest common divisor of φ and of u , which will necessarily be odd and not divisible by a , and put $\varphi = m\varphi'$, $u = mu'$, values whose substitution in the equation obtained above changes it into this one:

$$(t + \psi)(t - \psi) = m^2(\varphi'^2 + au'^2).$$

This equation shows that $t + \psi$ is composed of two factors, one of which divides m^2 , the other $\varphi'^2 + au'^2$, and that the same holds for $t - \psi$. If we denote the factors of $t + \psi$ by E and K and those of $t - \psi$ by F and L , we shall have:

$$\begin{aligned} t + \psi &= EK, & t - \psi &= FL, \\ m^2 &= EF, & \varphi'^2 + au'^2 &= KL. \end{aligned}$$

One will ascertain, as in paragraph 4, that the numbers E and F are relatively prime. It follows from this and from the equation $m^2 = EF$ that each of these numbers is a square, so that:

$$\left(\frac{E}{a}\right) = 1, \quad \left(\frac{F}{a}\right) = 1.$$

The difference of the odd numbers $t + \psi$ and $t - \psi$ being 2ψ and consequently divisible by 4, one sees that these two numbers are either both of the form $4n + 1$, or both of the form $4n + 3$. Since, on the other hand, the numbers E and F are both of the form $4n + 1$ (these numbers being odd squares), it follows from the equations $t + \psi = EK$, $t - \psi = FL$, that the numbers K and L are either both of the form $4n + 1$ or both of the form $4n + 3$, and that the first or the second of these cases holds according as the numbers $t + \psi$ and $t - \psi$ are both of the form $4n + 1$, or both of the form $4n + 3$. Let us now distinguish two cases, according as p is of the form $8n + 1$ or of this one: $8n + 5$. One sees, from the equation $p^2 = \varphi^2 + \psi^2$, that, in the first case, ψ is divisible by 4, whence it follows that $t + \psi$ and $t - \psi$ are then both of the form $4n + 1$ or both of the form $4n + 3$, according as t is of the form $4n + 1$, or of this one: $4n + 3$. The contrary holds in the case of $p = 8n + 5$, and the results relating to these two cases can be comprised in this statement:

The numbers $t + \psi$ and $t - \psi$, and consequently also the numbers K and L , are both of the form $4n + 1$ or of this one: $4n + 3$, according as $(-1)^{\frac{p-1}{4} + \frac{t-1}{2}}$ is equal to 1 or to -1 .

It follows from the equation $\varphi'^2 + au'^2 = KL$ that the numbers K and L are divisors of the formula $\varphi'^2 + au'^2$ (where φ' and au' are relatively prime and where a denotes a prime number $4n + 1$). Now, one knows that, if R denotes an odd divisor of such a formula, one has:

$$\left(\frac{R}{a}\right) = 1, \quad \text{or} \quad \left(\frac{R}{a}\right) = -1,$$

according as R is of the form $4n + 1$ or of the form $4n + 3$. (*Theory of Numbers*, no. 196.)

Applying this theorem to the numbers K and L , one obtains:

$$\left(\frac{K}{a}\right) = \left(\frac{L}{a}\right) = (-1)^{\frac{p-1}{4} + \frac{t-1}{2}},$$

whence one will then conclude, by multiplying by the expressions $\left(\frac{E}{a}\right) = 1$, $\left(\frac{F}{a}\right) = 1$, found above:

$$\left(\frac{t + \psi}{a}\right) = \left(\frac{t - \psi}{a}\right) = (-1)^{\frac{p-1}{4} + \frac{t-1}{2}},$$

or, which comes to the same thing, a being a prime number $4n + 1$:

$$\left(\frac{\psi + t}{a}\right) = \left(\frac{\psi - t}{a}\right) = (-1)^{\frac{p-1}{4} + \frac{t-1}{2}}.$$

If one compares what has just been proved with the result (θ) , obtained above, one will see that a is or is not a biquadratic residue with respect to p , according as one has:

$$\left(\frac{\psi+t}{a}\right) = \left(\frac{\psi-t}{a}\right) = \left(\frac{t}{a}\right),$$

or:

$$\left(\frac{\psi+t}{a}\right) = \left(\frac{\psi-t}{a}\right) = -\left(\frac{t}{a}\right).$$

The equation $p = t^2 - au^2$ shows that, if one denotes by χ either of the two roots of the congruence $\chi^2 \equiv p \pmod{a}$, one has $t \equiv \chi$ or $t \equiv -\chi \pmod{a}$. It follows from this, a being of the form $4n+1$, that $\left(\frac{t}{a}\right) = \left(\frac{\chi}{a}\right)$. On the other hand, since by virtue of what precedes $\left(\frac{\psi+t}{a}\right) = \left(\frac{\psi-t}{a}\right)$, one sees that one always has:

$$\left(\frac{\psi+\chi}{a}\right) = \left(\frac{\psi+t}{a}\right) = \left(\frac{\psi-t}{a}\right).$$

One concludes from this, having regard to what was said only a moment ago, that the number a will be a biquadratic residue with respect to p when one has $\left(\frac{\chi+\psi}{a}\right) = \left(\frac{\chi}{a}\right)$ or, which is the same thing, when $\left(\frac{\chi(\chi+\psi)}{a}\right) = 1$, and that a will be a biquadratic non-residue with respect to p when one has $\left(\frac{\chi+\psi}{a}\right) = -\left(\frac{\chi}{a}\right)$, or, which comes to the same thing, when $\left(\frac{\chi(\chi+\psi)}{a}\right) = -1$.

This result concerns the case in which φ is not divisible by a . It remains to examine the case of φ divisible by a . Treating this case in a similar way, one finds that, when it holds, one of the numbers $t+\psi$, $t-\psi$ (we shall denote it by $t \pm \psi$) is divisible by a and that, as in the first case, the other, $t \mp \psi$, is such that:

$$\left(\frac{t \mp \psi}{a}\right) = (-1)^{\frac{p-1}{4} + \frac{t-1}{2}},$$

or, which comes to the same thing, on adding to $t \mp \psi$ the number $t \pm \psi$, a multiple of a :

$$\left(\frac{2t}{a}\right) = (-1)^{\frac{p-1}{4} + \frac{t-1}{2}}.$$

Comparing this with what was proved above (θ) , one will find that, in the case of φ divisible by a , a is or is not a biquadratic residue with respect to p , according as one has $\left(\frac{2}{a}\right) = 1$, or $\left(\frac{2}{a}\right) = -1$, or, which is the

same thing, by a known theorem, according as a is of the form $8n + 1$ or of the form $8n + 5$.

The results we have just obtained stand as do those that we established in paragraph 4; they still hold even when p cannot be put under the form $t^2 - au^2$, and presuppose only that the prime numbers $a = 4n + 1$, and $p = 4n + 1$, are such that $\left(\frac{a}{p}\right) = 1$. To prove them in this extension, it will suffice to follow the procedure indicated towards the end of paragraph 4, and to rely on the auxiliary proposition discussed there, whose statement was given the generality necessary for it to serve as a basis, at the same time, for the demonstration of the general theorem at the end of paragraph 4 and for that of the theorem we are about to state. These two theorems properly form only one, and by proceeding as was said in the cited place, that is, by applying to the equation $t^2 \pm Mu^2 = p$, where M denotes the product of any number of distinct prime numbers, considerations analogous to those employed in this paragraph and in the preceding ones, one will demonstrate these two theorems simultaneously, or rather one will arrive at a theorem in whose statement they are united.

Theorem II.

“ a denoting a prime number $4n + 1$, and p another prime number $4n + 1$, such that $\left(\frac{a}{p}\right) = 1$, if one puts $p = \varphi^2 + \psi^2$ (where ψ is supposed even), one will be able to decide in the following way whether a is or is not a biquadratic residue with respect to p . If φ is divisible by a , a will or will not be a biquadratic residue with respect to p , according as a is of the form $8n + 1$ or of this one: $8n + 5$. If φ is not divisible by a , one will seek a number χ such that $\chi^2 \equiv p \pmod{a}$. This being done, a will or will not be a biquadratic residue with respect to p , according as one has $\left(\frac{\chi(\chi+\psi)}{a}\right) = 1$ or $\left(\frac{\chi(\chi+\psi)}{a}\right) = -1$.”

By successively taking $a = 5$, $a = 13$, one will have the following particular theorems, which must be regarded as rigorously proved by what precedes, the formula $t^2 - au^2$ having, for these values of a , no quadratic divisor other than this one: $t^2 - au^2$, or in other words, every divisor of $t^2 - au^2$ being itself of the form $t^2 - au^2$.

“ p denoting a prime number of one of the forms $20n + 1$, $20n + 9$, if one puts $p = \varphi^2 + \psi^2$ (where ψ is supposed even) one readily ascertains that one of the numbers φ , ψ is a multiple of 5. This being done, I say that 5 will or will not be a biquadratic residue with respect to p , according as it is ψ or φ that is divisible by 5.”

etc.

In indicating above a method that could serve to establish theorems I and II in all their generality, we said that, in following this method, one found oneself obliged to rely on a subsidiary proposition that appears rather difficult to demonstrate with complete rigour. On reflecting on this difficulty, I recognized that it was very easy to remedy it by modifying slightly the procedure outlined above. This modification is extremely slight and consists in this: instead of considering the equations $t^2 - bu^2 = p$, $t^2 - au^2 = p$, which may fail to be possible although one has $\left(\frac{b}{p}\right) = 1$, $\left(\frac{a}{p}\right) = 1$, one must apply considerations of the preceding kind to the more general equations $t^2 - bu^2 = ps^2$, $t^2 - au^2 = ps^2$ (one advantageously replaces the latter by this one: $t^2 + au^2 = ps$, which can be treated more simply), which can always be satisfied when the conditions $\left(\frac{b}{p}\right) = 1$, $\left(\frac{a}{p}\right) = 1$ hold, as follows from the beautiful theorem that Mr. LEGENDRE gave for deciding the possibility or impossibility of the equation $\alpha x^2 + \beta y^2 = \gamma z^2$ (*Theory of Numbers*, no. 27).

The demonstration thus modified, besides being entirely rigorous, has the further advantage of greater simplicity, as any reader who has fully grasped the spirit of the preceding considerations will be able to judge by developing this demonstration according to the indication just given.

Addition to the preceding memoir.

Although the indication that we gave at the end of the preceding memoir concerning the procedure to follow in order to establish in an entirely rigorous way theorems I and II of our memoir may suffice for any reader familiar with indeterminate analysis, we believe we are not doing something useless by developing in detail, in this addition, the demonstrations of the theorems in question, these demonstrations being susceptible of being greatly

simplified by means of a few considerations that may not immediately present themselves to the mind. We begin with the demonstration of the first of the theorems cited.

Let us denote by p a prime number $4n + 1$, by b a prime number $4n + 3$, such that $\left(\frac{b}{p}\right) = 1$, and consider the equation:

$$(\alpha') \quad t^2 - bu^2 = ps^2.$$

By the theorem already cited (*Theory of Numbers*, no. 27), the necessary conditions for this equation to be possible are these: $\left(\frac{b}{p}\right) = 1$ and $\left(\frac{p}{b}\right) = 1$. Now, of these conditions the first holds by hypothesis, and as for the second, it follows from the theorem known as the law of reciprocity that it always holds when the first is satisfied. The equation is therefore soluble. It is evident that one may suppose the numbers t, u, s that satisfy it to be relatively prime when compared two at a time. The equation being in this state, one of these three numbers will be even, the other two odd, and one readily ascertains that s cannot be even. The number s being odd, of the numbers t, u , one will be even, the other odd, and there would be two cases to examine according as t is even or odd. But although these two cases can be treated by the same analysis, it is simpler to consider only one of them, that of t odd for example, and to show that the other can easily be reduced to it. Let us therefore show that it is always permissible to suppose t odd in the equation $t^2 - bu^2 = ps^2$. If t were even in the equation (α') , one would deduce from the numbers t, u , by the means indicated in the note relating to the beginning of paragraph 3 of the preceding memoir, other numbers t' (odd), u' (even) such that $t'^2 - bu'^2 = ps^2$. We may therefore regard t as odd in all that follows.

Let us decompose the numbers t and u into their simple factors, putting:

$$t = ll'l'' \dots, \quad u = 2^\nu kk'k'' \dots,$$

$l, l', l'', \dots$ and $k, k', k'', \dots$ denoting odd prime numbers. Mere inspection of the equation (α') gives:

$$\left(\frac{p}{k}\right) = 1, \quad \left(\frac{p}{k'}\right) = 1, \quad \left(\frac{p}{k''}\right) = 1, \quad \dots$$

One will then have, on applying the law of reciprocity, p being a prime number $4n + 1$:

$$\left(\frac{k}{p}\right) = 1, \quad \left(\frac{k'}{p}\right) = 1, \quad \left(\frac{k''}{p}\right) = 1, \quad \dots$$